

does not address the underlying pathophysiology of aHUS and has limited effects on long-term prognosis for these patients (Raina, et al. 2019). Furthermore, PE/PI pose many challenges, such as adverse effects mainly related to transfusion reactions, access site infections as well as burden to patients with continued therapy (Nester and Thomas 2012). Because FH, FI, FB, and C3 are synthesized in the liver, liver transplantation remains an alternative option for some patients to provide a source of normal protein (Nester and Thomas 2012). Patients reaching kidney failure require kidney transplant or combined liver-kidney transplantation. aHUS patients have high rates of disease recurrence associated with graft loss after kidney transplantation with the outcome being worse in patients with FH and FI mutations (Saland J M, et al. (2009) Liver-kidney transplantation to cure atypical hemolytic uremic syndrome. *J Am Soc Nephrol*; 20:940-9).

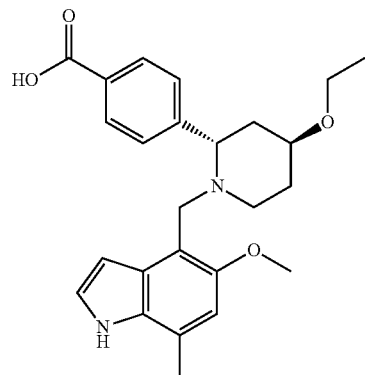
[0007] LNP023 (iptacopan) is a novel, orally administered, small molecular weight, first-in-class, selective protease inhibitor that binds to FB (Bb domain) and inhibits C3- (i.e., C3bBb) and C5 (C3bBbC3b) convertases thereby blocking the formation of the MAC. In addition, iptacopan blocks the amplification phase, and halts the complement activation process. In multiple in vitro and in vivo non-clinical mechanistic studies, iptacopan has demonstrated inhibition of the AP (Schubart A, et al. (2019) Small-molecule factor B inhibitor for the treatment of complement-mediated diseases. *Proc Natl Acad Sci USA*; 116(16):7926-31). Based on the well-characterized AP dysregulation within aHUS pathophysiology, supported by approved terminal complement inhibitor therapies for treatment of aHUS (eculizumab and ravulizumab); an AP pathway inhibitor such as iptacopan is likely to provide therapeutic benefit in aHUS patients.

SUMMARY

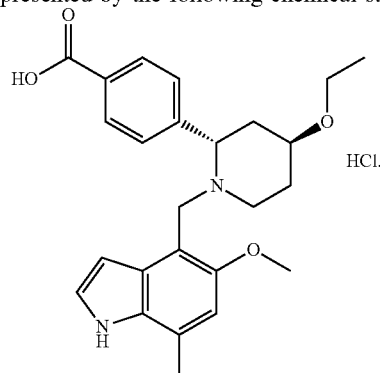
[0008] The disclosure relates to methods of treating complement driven diseases, and in particular, atypical hemolytic uremic syndrome (aHUS), with LNP023 (iptacopan) or a pharmaceutically acceptable salt thereof, e.g. iptacopan hydrochloride. Further, the disclosure relates to a Phase 3 clinical study to determine safety and efficacy of iptacopan or a pharmaceutically acceptable salt thereof, e.g., iptacopan hydrochloride, in patients with aHUS. In an embodiment, the patients are treatment naive to complement inhibitor therapy, including anti-C5 antibody therapy. In an embodiment, the patients have been previously treated, or are currently being treated, with complement inhibitor therapy, immunosuppressive therapy, or other therapy prescribed for the treatment of aHUS. Iptacopan belongs to the class of Factor B inhibitors of the complement pathway and acts by inhibiting or suppressing the amplification of the complement system caused by C3 activation irrespective of the initial mechanism of activation.

[0009] The disclosure also relates to pharmaceutical compositions, uses, kits, etc. related to iptacopan or a pharmaceutically acceptable salt thereof, e.g., iptacopan hydrochloride.

[0010] Iptacopan is chemically designated as 4-((2S,4S)-(4-ethoxy-1-((5-methoxy-7-methyl-1H-indol-4-yl)methyl)piperidin-2-yl))benzoic acid and can be represented by the following chemical structure:



[0011] Iptacopan hydrochloride is chemically designated as 4-((2S,4S)-(4-ethoxy-1-((5-methoxy-7-methyl-1H-indol-4-yl)methyl)piperidin-2-yl))benzoic acid hydrochloride and can be represented by the following chemical structure:



[0012] Iptacopan, iptacopan hydrochloride, and methods of preparation are disclosed in U.S. Pat. Nos. 9,682,968 and 10,093,663 (see Examples 26a, 26c and 26d), which are incorporated herein by reference in their entirety.

[0013] The form of iptacopan hydrochloride used as the investigational study drug for this Phase 3 study is a monohydrate (Form HB) as shown in the formula below:

